Bosen Yang

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EDUCATION

University of California, Los Angeles (Cum Laude, 3.937) Degree Conferred – Jun 2026

- *B.S. in Applied Mathematics (Computing Specialization), B.S. in Statistics and Data Science*
- Core Coursework: Real/Complex Analysis, Numerical Analysis, Math Modeling, Linear Algebra, Diff Eqn, Graph Theory, OOP, Software Tools, Data Structure, Prob Theory, Comp/Math Stats, Linear Model, Exp Design, Data Mining, Optimization, Monte Carlo, Consulting.

University of Michigan, Ann Arbor Degree Expected – Dec 2027

- *M.S. in Applied Statistics*

EXPERIENCE

Research Assistant *UCLA Trustworthy AI Lab – Directed by Professor Cheng* Los Angeles, CA; Apr 2026 – Present

- Simulated **interactive benchmark games for LLM agents** such as prisoner's dilemma with communication in Python. Conducted analysis of multi-agent trust-building, betrayal, decision-making with and without conversation from Stats 263 class game response data.

Directed Reading *UCLA Mathematics – Mentored by PhD Candidate William Chang* Los Angeles, CA; Jan 2026 – Present

- Investigated theoretical foundations of **online learning** for sign-activation **neural networks**.
- Investigated margin-preserving depth reduction to eliminate depth dependence in existing neural network mistake bounds. Analyzed polyhedral representations of threshold networks and established margin preservation for collapsed two-layer architectures.

Directed Reading *UCLA Stats 199 – Mentored by Professor Michailidis* Los Angeles, CA; Apr 2026 – June 2026

- One-on-one simulation study on **high-dimensional statistical estimation under network dependence in Bayesian perspective**.
- Implemented data simulation with Gibb's sampling and evaluated estimation performance under different priors in Python.

Data Science / ML Intern *Jietai Solar* Anhui, Chuzhou, CN; June 2024 – Aug 2024

- Analyzed photovoltaic product sales and production cost data for systematic manufacturing schedule optimization. Adopted time-series forecasting and regression modeling techniques in R and Python to model short to medium term demand.
- Queried and imported historical production volume and profit/loss data using SQL, and constructed forecasting frameworks informed by cost accounting and market trends.
- Implemented Machine Learning algorithms (RFR, LSTM etc.) with mlr3 and scikit-learn. Performed cross validation and scenario analysis to improve forecast stability and accuracy on rolling basis.

Instructional Assistant & Reader / Grader *UC San Diego Mathematics* San Diego, CA; Sep 2023 – Dec 2023

- Course: MATH 20E Vector Calculus - Holding 4 discussion sections and weekly office hours; grading homework/exams; proctoring exams.

PROJECTS

Multi-Sensor Temperature - Enzyme Activity Experiment & Data Denoising July – Aug 2025

- Designed and built a multi-sensor temperature monitoring module using Arduino UNO controller to study ALP enzyme solution.
- Simulated heat diffusion via the fundamental solution of the heat equation, yielding 3D Gaussian heat distribution. Designed a monotonic heating experiment with delayed enzyme reaction modeling and applied method of char. to approximate optimal ALP activity temperature.
- Designed an integrated real-time denoising framework combining Dynamic Inverse Variance Weighting, FFT, and Kalman Filter in R. monitoring and control within the designated interval of the module.

International Global Terms of Trade Time Series Forecasting Aug – Sep 2025

- Analyzed Terms of Trade (ToT) time series (CN, JP, IN, AU, US) and constructed multivariate forecasting models in R
- Implemented and compared ARIMA, ETS (with Denton-Cholette Method), Multivariate, and GARCH model using forecast, vars, and rugarch in R, with stationarity and normality tests, Box-cox transformation and correlation analysis to inform model selection.
- Conducted trend analysis of ToT and derived econometric insights to inform business decision-making and macroeconomic forecasting.

SELECTED REVIEWS

[1] Yang, Bosen. "Application of matrix decomposition in machine learning." *2021 IEEE International Conference on Computer Science, Electronic Information Engineering and Intelligent Control Technology (CEI)*. IEEE, 2021.

[2] Yang, Bosen. "Internal Combustion Engine Vehicles Fuel Efficiency Regression Analysis with Physical and Engineering Perspective." *Highlights in Business, Economics and Management* 65 (2025): 382-392.

SKILLS

- **Programming:** R, Python, MATLAB, Java
- **Software Tools & Frameworks:** LaTeX, SQL, VS Code, Git, Word/Excel